Zerodha (*A Stock Trading Platform Database System*)

🏢 Company Overview

Zerodha officially started its operations on **15th August 2010** with the objective of removing barriers faced by traders and investors in terms of cost, technology, and support. It was founded by **Nithin Kamath, Nikhil Kamath, and Austin Prakesh**.

The name *Zerodha* is derived from the combination of **"Zero"** and **"Rodha"** (a Sanskrit word meaning barrier), symbolizing the company's mission to eliminate obstacles in trading and provide a seamless investment experience.

Zerodha offers a range of technology-driven platforms, including:

- **Kite** – Trading platform for buying and selling securities
- **Coin** – Platform for direct mutual fund investments
- **Varsity** – Educational platform for learning about markets
- **Kite Connect API** – API for developers to build trading applications

The company has approximately **7.9 to 8.1 million active users** and is known for its simplicity, transparency, and innovative approach.

Additionally, Zerodha provides access to partner platforms such as:

- **Smallcase** – Thematic investing platform
- **Tijori** – Investment research platform
- **Streak** – Algorithmic trading platform
- **Sensibull** – Options trading platform
- **Ditto** – Insurance platform

🚀 Product Dissection & Real-World Problems Solved

Zerodha has transformed the Indian brokerage industry by addressing key challenges faced by retail investors. It focuses on reducing costs, simplifying trading platforms, and improving financial literacy.

🔴 Problem 1: High Brokerage Fees

Challenge: Traditional brokers charge high commissions, making trading expensive for retail investors.

Solution : Zerodha introduced a discount brokerage model, charging:

- Introduced **discount brokerage (₹0 delivery & 0.03% or 20 rs for intraday & Flat ₹20 per trade for F&O)**
- Made trading affordable for retail investors
- This drastically reduced trading costs and made investing accessible.

🔴 Problem 2: Complex Trading Platforms

Challenge: Most trading platforms are difficult to use, especially for beginners.

Solution: Zerodha developed **Kite**, a clean and user-friendly interface that:

- Simplifies buying and selling of stocks
- Provides real-time charts with advanced technical indicators
- Works seamlessly on both web and mobile platforms
- Offers advanced charting tools for technical analysis
- Enables fast order execution with minimal latency
- Includes watchlists for tracking selected stocks
- Provides portfolio tracking and profit & loss analysis
- Supports multiple order types like market, limit, and stop-loss
- Offers a customizable dashboard for better usability
- Ensures security with two-factor authentication

🔴 Problem 3: Lack of Financial Education

Challenge: Many users don’t understand stock markets, leading to poor decisions.

Solution: Zerodha created Varsity by Zerodha, which offers:

- Zerodha Varsity offers **structured learning paths** designed like chapters and modules
- Includes **quizzes and assessments** after each module to test understanding
- Provides **real-world market examples and case studies** for practical learning
- Available as a **mobile app for learning on the go**
- Covers topics like **technical analysis, fundamental analysis, derivatives, and risk management**
- Content is **regularly updated** according to market changes
- Supports **self-paced learning**, suitable for students and working professionals
- Offers **certification (optional learning proof)** for completed modules

🔴Problem 4: Limited Access to Investments

Challenge : Investing in mutual funds involved middlemen and commissions.

Solution : Zerodha introduced Coin, a platform for direct mutual fund investments. Initially priced at ₹50/month, it was later made completely free. Coin eliminates commissions, helping users achieve higher returns. It also allows investors to apply for new fund offers (NFOs) through both mobile and web platforms, providing a seamless and transparent investment experience. Provides an **auto-debit (mandate) feature** for SIPs, enabling automatic payments and ensuring users do not miss their investments.

🔴Problem 5: Portfolio Tracking & Analytics

Real-World Challenge : Users face difficulties in tracking investment performance, analyzing returns, and managing tax-related aspects.

Solution : Zerodha addresses this through Console, a comprehensive reporting and analytics platform. It provides profit and loss reports, detailed tax statements, and portfolio tracking. Additionally, it supports tax-loss harvesting to optimize taxes and allows users to participate in corporate actions like offer for sale (OFS) and buybacks. Console acts as a centralized dashboard for all investment insights and analytics.

🔴Problem 6: Manual Trading & Lack of Real-Time Data

Challenge : Traditional trading systems require users to manually monitor charts and place orders while also lacking easy access to real-time market data, leading to time consumption, missed opportunities, emotional decision-making, and difficulty in building automated trading systems.

Solution: Zerodha provides **Kite Connect**, which enables automated trading through APIs and offers WebSocket-based real-time market data (including price updates and OHLC), allowing users to build trading bots, execute orders faster, and make data-driven decisions.

📌 Top Features of Zerodha

User Accounts : Zerodha allows users to create trading accounts by providing personal and financial details such as name, email, PAN number, and bank information. These accounts serve as the foundation for all trading and investment activities, enabling users to securely access financial markets and manage their portfolios.

Trading Platform (Kite) : A core feature of Zerodha is its flagship platform, Kite, which allows users to buy and sell stocks, commodities, and derivatives. It provides real-time market data, advanced charting tools, and a user-friendly interface that simplifies the trading experience for both beginners and experienced investors.

Orders & Execution : Zerodha enables users to place different types of orders such as market orders, limit orders, GTT (Good Till Trigger) and stop-loss orders. Kites also provide the feature of alert of stocks. The platform ensures fast and efficient order execution, allowing users to take advantage of market opportunities with minimal delay.

Portfolio Management : Zerodha provides users with a comprehensive view of their investments through portfolio tracking features. Users can monitor their holdings, view profit and loss statements, and analyze performance over time, helping them make informed investment decisions.

Funds & Transactions : The platform allows seamless fund management, including deposits and withdrawals. Users can easily add funds to their accounts for trading and withdraw profits, ensuring smooth financial transactions within the system.

Mutual Fund Investment (Coin) : Through Zerodha Coin, Zerodha enables users to invest in direct mutual funds without any commission. This feature allows investors to maximize returns and diversify their portfolios efficiently.

Learning Platform (Varsity) : Zerodha supports user education through Zerodha Varsity, which provides structured and easy-to-understand content on trading and investing. It helps beginners gain knowledge and build confidence in financial markets.

API Integration (Kite Connect) : Zerodha provides Kite Connect API, which allows developers to build custom trading applications and automate trading strategies. This feature enhances flexibility and enables advanced users to create personalized trading solutions.

Reports & Backoffice (Console) : Zerodha offers a backoffice platform called Zerodha Console, where users can access detailed reports such as trade history, ledger balance, tax reports, and profit and loss statements. This ensures transparency and helps users manage their financial records effectively.

🗂️ Schema Design (Database Design)

This section defines the database schema for the Zerodha platform, capturing users, trading activities, and financial transactions.

👤 **User Entity :** Users are the core of the system. This entity stores all user-related information.

- **UserID (PK):** Unique identifier assigned to each user
- **Name:** Full name as per official records
- **Email:** Used for login and communication
- **Phone:** Registered mobile number for verification
- **PAN_Number:** Unique PAN for KYC compliance
- **Account_Type:** Type of account (Individual, NRI, HUF, Company etc.)
- **Registration_Date:** Date of account creation

💼 **Account Entity :** This entity stores details about the user's trading/demat account.

- **AccountID (PK):** Unique Demat/Trading account number
- **UserID (FK):** Links account to the user
- **Balance:** Available trading balance
- **Account_Status:** Active / Inactive / Dormant

📊 **Stock Entity :** This entity contains details about stocks available for trading.

- **StockID (PK):** Unique stock identifier
- **Company_Name:** Name of issuing company
- **Symbol:** Trading symbol (e.g., RELIANCE, TCS)
- **Market_Price:** Current stock price
- **Exchange:** NSE / BSE
- **Sector:** Industry classification
- **ISIN:** International identifier
- **52_Week_High / 52_Week_Low:** Performance indicators

- **Volume:** Trading volume
- **Last_Updated:** Price update timestamp

📈 **Order Entity :** This entity stores details of orders placed by users.

- **OrderID (PK):** Unique order identifier
- **UserID (FK):** User placing the order
- **StockID (FK):** Stock being traded
- **AccountID (FK):** Specifies which account is used to place the order
- **Order_Type:** Buy / Sell
- **Quantity:** Number of shares
- **Price:** Order price
- **Order_Date:** Timestamp of order placement
- **Status:** Pending / Executed / Cancelled
- **Order_Mode:** Market / Limit / Stop-Loss
- **Brokerage:** Charges applied
- **Exchange:** NSE / BSE
- **Order_Validity:** Day / IOC
- **Trigger_Price:** For stop-loss orders

💰 **Trade Entity :** This entity records executed trades.

- **TradeID (PK):** Unique trade identifier
- **OrderID (FK):** Associated order
- **Execution_Price:** Final trade price
- **Execution_Time:** Execution timestamp
- **Executed_Quantity:** Quantity fulfilled
- **Trade_Type:** Intraday / Delivery
- **Charges:** Fees and taxes
- **Net_Amount:** Final transaction value

📉 **Portfolio Entity :** This entity tracks user holdings and investment performance.

- **PortfolioID (PK):** Unique portfolio record ID
- **UserID (FK):** Portfolio owner
- **StockID (FK):** Stock held
- **Quantity_Held:** Total shares owned
- **Average_Price:** Average purchase price

- **Current_Price:** Latest market price
- **Total_Investment:** Quantity × Avg Price
- **Current_Value:** Quantity × Current Price
- **Unrealized_PnL:** Profit/Loss (not booked)
- **Last_Updated:** Last update timestamp
- **UNIQUE(UserID, StockID):** Ensures that a user cannot have duplicate entries for the same stock in the portfolio.

💳 **Transaction Entity :** This entity stores financial transactions of users.

- **TransactionID (PK):** Unique transaction identifier
- **UserID (FK):** User performing transaction
- **AccountID (FK):** Unique Demat/Trading account number
- **Transaction_Type:** Deposit / Withdrawal / Buy / Sell
- **Amount:** Transaction amount
- **Transaction_Type:** Deposit / Withdrawal
- **Date:** Transaction timestamp
- **Transaction_Status:** Successful / Pending / Failed
- **Payment_Mode:** UPI / Net Banking / Bank Transfer
- **Reference_ID:** External transaction reference
- **Balance_After_Transaction:** Updated balance
- **Remarks:** Additional notes

🔗 Relationships

Users → Accounts (1 : N) – Each user can have multiple accounts (e.g., trading, demat), and each account belongs to one user.

Users → Orders (1 : N) – Each user can place multiple buy/sell orders, and each order belongs to one user.

Accounts → Orders (1 : N) - Each account can have multiple orders, and each order is placed from one account.

Stocks → Orders (1 : N) - A stock can have multiple orders, and each order is associated with one stock.

Orders → Trades (1 : N) – Each order can result in one or multiple trades (due to partial execution).

Users → Portfolio (1 : N) – Each user can have multiple portfolio records (different stocks).

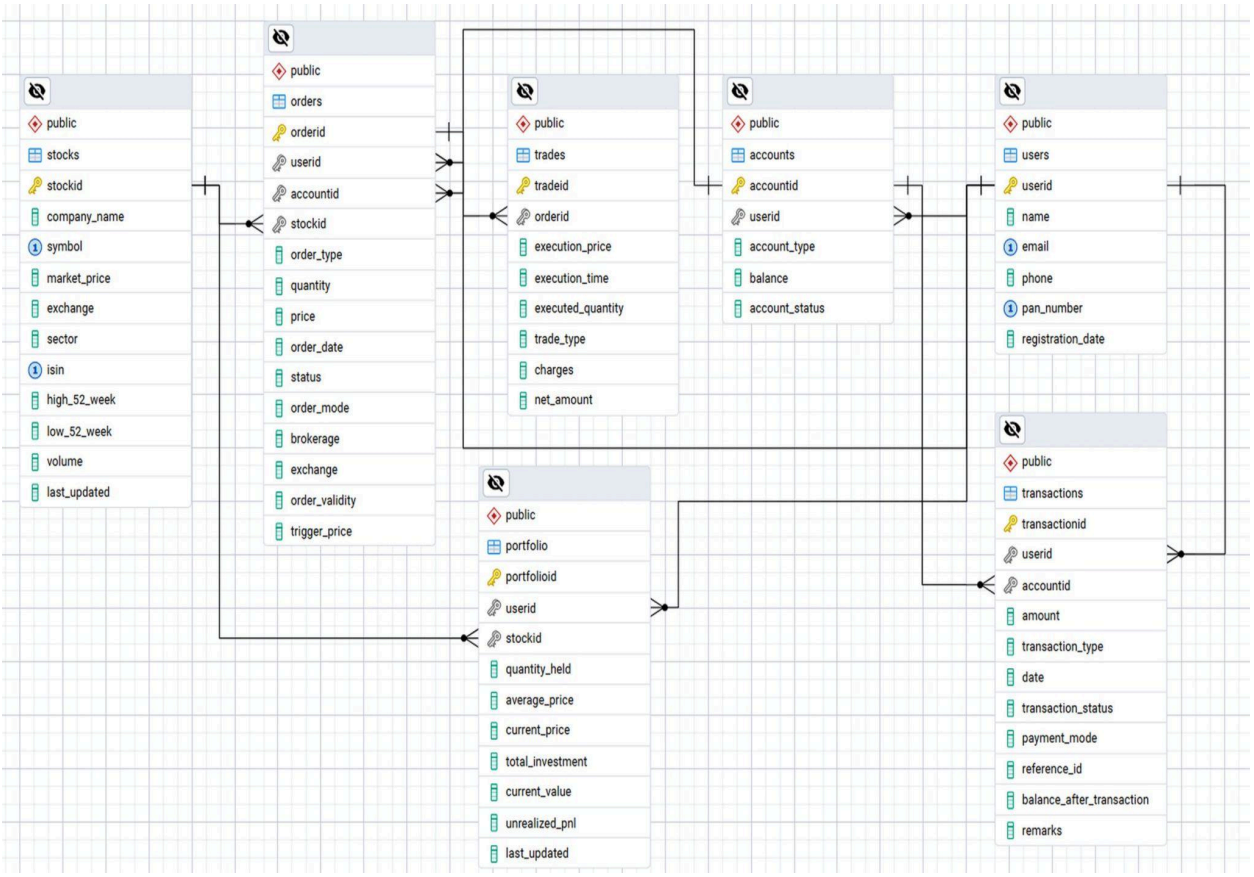
Stocks → Portfolio (1 : N) – A stock can be held by multiple users, and each portfolio entry corresponds to one stock.

Users → Transactions (1 : N) – Each user can have multiple transactions like deposits and withdrawals.

Accounts → Transactions (1 : N) – Each account can have multiple transactions, and each transaction belongs to one account.

Portfolio Constraint - A user can hold a stock only once in the portfolio.

📊 ER Diagram



✅ Conclusion

Zerodha has transformed India's trading ecosystem by:

- Making investing affordable
- Simplifying trading platforms
- Educating users
- Eliminating intermediaries

This project successfully designed and implemented a relational database system for a stock trading platform similar to Zerodha. It demonstrates how real-world trading operations such as user management, order placement, trade execution, portfolio tracking, and financial transactions can be efficiently handled using a well-structured database. The schema ensures data integrity, scalability, and optimized performance, making it suitable for handling large volumes of trading data.